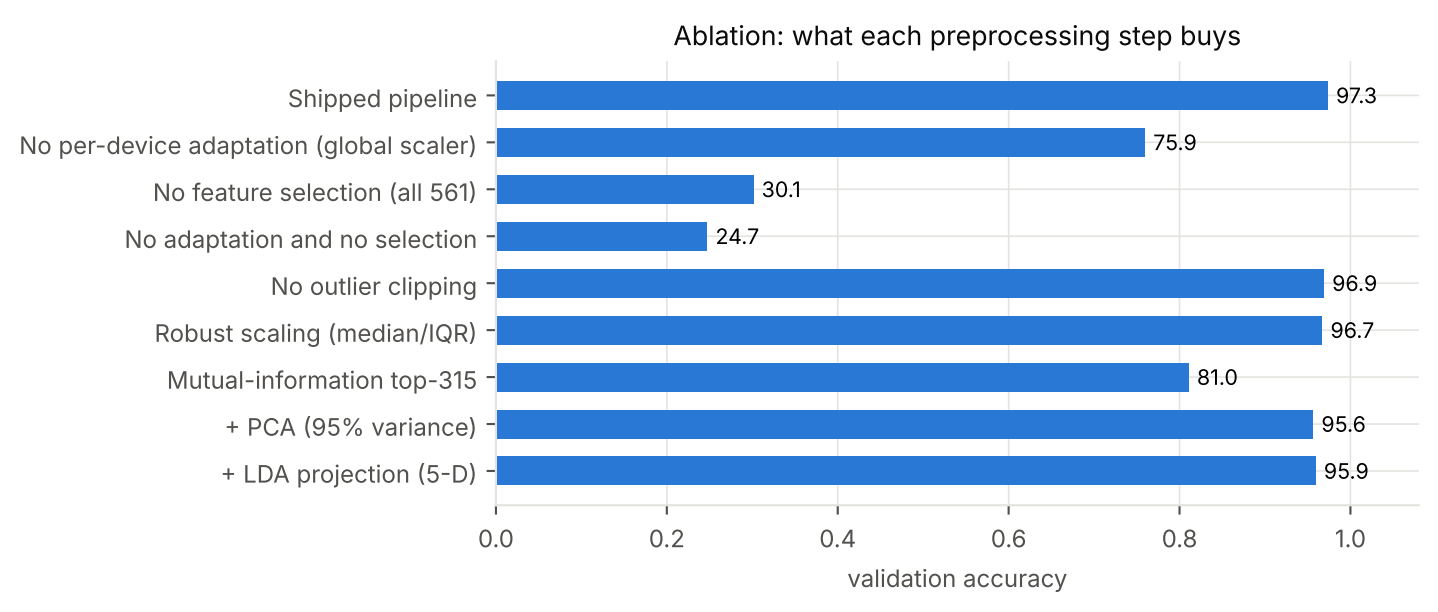
Ablation: what each preprocessing step buys
Shipped pipeline
No per-device adaptation (global scaler)
No feature selection (all 561)
No adaptation and no selection
No outlier clipping
Robust scaling (median/IQR)
Mutual-information top-315
+ PCA (95% variance)
+ LDA projection (5-D)
97.3
75.9
30.1
24.7
96.9
96.7
81.0
95.6
95.9
0.0
0.2
0.4
0.6
0.8
1.0
validation accuracy